

MamKanformer: A hybrid Mamba and KAN model based on Medformer for Medical Time-Series Classification

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Abstract—The rapid advancements in sensing technology and artificial intelligence have spurred a substantial number of novel demands and challenges within the realm of medical time series analysis. Notably, the effective extraction and utilization of the long-term and short-term dependency relationships, along with the features at various granularities encapsulated within high-dimensional and multivariate time series, have emerged as one of the pressing issues that demand immediate resolution. To address these, we propose a deep learning model MamKanformer for multichannel EEG signal processing. The model firstly solves the problem of long-short term dependencies which are difficult to be modeled effectively by Feed Forward in Transformer by using Mamba architecture. Subsequently, Kan is utilized to capture the critical feature information of different granularities contained in the time series data to assist in enhancing the output of the model. Finally, extensive experiments conducted on three public datasets have effectively verified the efficacy of the method proposed in this paper. Specifically, the MamKanformer model has demonstrated a superior performance over the other 11 baselines. It has achieved an average top ranking across all six metrics within the three datasets, which unequivocally attests to the outstanding capabilities of our proposed method.

Index Terms—EEG, multivariate time series, Transformer, Mamba, Kan

I. INTRODUCTION

Electroencephalography (EEG) data is a type of medical time-series data used to record the electrical activity of the brain. It measures the electrical activity of neurons in the cerebral cortex by placing electrodes on the scalp. These electrodes capture the tiny electrical signals emitted by the neurons and convert them into graphical form to show the pattern and frequency of brain waves, which is widely used in the fields of medical treatment, brain-computer interface,

emotion recognition and neurofeedback. In practice, to capture the subtle changes in neural activity, the sampling rate of EEG data is generally high, reaching 250 to 1000 Hz, so the EEG signals are usually characterized by low intensity, high dimensionality, high frequency components, and easily disturbed by noise, which brings a lot of challenges to the processing of EEG data.

With the rapid development of artificial intelligence technology, more and more machine learning models are used in medical time series classification, most of which are based on manual feature extraction or traditional neural network models, but these methods are not able to extract the features of long time series well. At present, in the field of time series and natural language, Transformer-based methods have achieved effective results, and have very good results in time series classification, prediction, and anomaly detection. However, for the classification of medical time series, Transformer is not fully suitable for this task, and has some limitations in capturing long-range dependencies and global features. The reason is that in this task context, Transformer integrates multiple inconsistent measurements with the same timestamp into a single marker. However, the physical meanings of these data points may vary greatly. As a result, it is difficult to effectively integrate the diverse information generated at the same point in time, which makes it difficult for the model to extract useful features from them, thus adversely affecting the overall performance. For example, while iTransformer [15] provides enhanced feature representation and robustness to variability, it is computationally intensive, requires large annotated datasets, and is prone to overfitting, posing challenges in terms of training and interpretability. Informer [16] networks

have difficulty along the time dimension to capture coarse-grained features. Fedformer [17] emphasizes scale-awareness to some extent by employing different computational paths for the trend and seasonal components of the input time series; however, this structural prior focuses on only two scales: the low and high frequency components.

In order to solve the above problems, this paper proposes the MamKanformer model, which is based on MedFormer [1], firstly, introduces the Mamba architecture to learn the long- and short-term dependencies in the time series data more efficiently, and then, utilizes the Kan model to capture the key information in the time series data to assist the model output, enhance the output of the model. The proposed model can efficiently learn the features within a specific granularity and the correlations between different granularities for EEG data processing.

In summary, our contributions are as follows:

- We propose a feature extraction model, MamKanformer, which significantly improves the performance in complex time series data by improving the feature extraction strategy.
- Mamba is introduced to make up for the shortcomings of the traditional feed-forward network, an innovation that not only optimizes the computational process, but also dramatically improves the computational efficiency of the model, making our model more practical for real-time applications.
- The feature processing was further optimized using the KAN-assisted Projection mechanism. This mechanism enhances the model's adaptability in different tasks and enables it to handle diverse data inputs more flexibly.

II. RELATED WORK

A. Medical Time Series

Medical time series is collected from the human body over time, which is mainly used for disease diagnosis and health monitoring. The main publicly available datasets include EEG [2], ECG (electrocardiogram) [3] and EOG (electrooculogram) [4]. Each type of data can provide different value to different aspects of healthcare. For example, EEG data is used to monitor seizures, ECG data is used to identify cardiac arrhythmias, and EOG data can be used to analyze eye movements and sleep quality. Existing work relies on recognizing biomarkers or deep learning models based on CNNs and Transformers. For example, EEGNet [5] is designed specifically for EEG signals and uses deeply separable convolution to extract time and frequency domain features with significant results. DeepConvLSTM [6] is capable of extracting both local and temporal features by combining convolutional layers and LSTM for EEG and other physiological signals.

B. Hybrid Mamba And Transformer

In recent work, combining the Mamba [7] and Transformer architectures has led to the development of hybrid models aimed at capturing local and global features more efficiently. One such model is TransMamba [8], which utilizes

the complementary strengths of the two architectures in tasks such as image recovery and deraining. The TransMamba network uses a two-branch encoder-decoder architecture, where one branch contains a Spectral Domain Transformer Block (SDTB) that focuses on long-range dependencies through Spectral Banded Self-Attention (SBSA). Another branch uses the spatial-domain Mamba layer, which efficiently handles local short-sequence modeling through techniques such as bi-directional state-space models (SSM [14]). By combining these two approaches, the model can capture both fine-grained local features and broader global patterns, which is particularly useful for tasks that require detailed and large-scale contextual understanding. In addition, another hybrid approach, Jamba [9], combines the Mamba layer and the Transformer attention layer, demonstrating how their complementary mechanisms can improve efficiency and performance. By mixing the layers in different proportions, the model can manage memory more efficiently, making it suitable for processing long sequences while keeping computational requirements low. These hybrid architectures highlight the flexibility and robustness that results from combining state-space models such as Mamba with the Transformer architecture, known for its ability to capture global remote dependencies.

C. KAN For Time Series

Recently, Kolmogorov-Arnold Networks (KAN) [10] have received a lot of attention as a different type from traditional multilayer perceptrons (MLPs). Several studies have explored the performance of KAN in time series classification, demonstrating its effectiveness in capturing time-dependent and nonlinear patterns [11]. For example, Cristian J et al. [12] applied KAN to the task of time series forecasting and classification, and there was a significant improvement in the forecasting accuracy. Another KANformer study outperforms traditional RNNs in terms of speed and accuracy by subjecting KAN to time series prediction in stocks, where its ability to approximate complex nonlinear relationships helps. The study by Chen et al. [13] combines KAN with self-attention layers to achieve state-of-the-art performance in fine-grained classification of bone marrow cells. These advances demonstrate that KAN is a versatile and powerful tool for modeling time series data, especially in applications where high dimensionality and noisy data pose a challenge to traditional methods.

III. METHOD

In this section, we first introduce the overall framework of the model. Next, we analyze the Mamba mechanism and finally discuss the KAN network.

A. Overview of MamKanformer

Figure.1 is the overall structure of our model. In our model, we mainly did two things. First, Mamba solved the problem that Feed Forward in the basic model Medformer was difficult to effectively model the long-short term dependencies in the sequence. Second, the KAN network was added to the output layer to assist the final output. Therefore, we can see

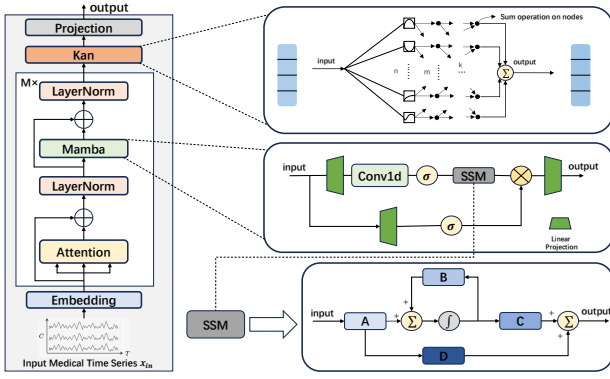


Fig. 1: Architecture of MamKanformer:Based on the structure of Medformer, replacement of Feed Forward module by Mamba and KAN network is added to the output layer.

that when $x_{in} \in \mathbb{R}^{T \times C}$ is input, if it passes through the position encoding and self-attention structure, the output will be $x_{mid} \in \mathbb{R}^{N \times D}$, Then in the Mamba layer, the following equation will be given [7]:

$$x_{out} = SSM(\sigma(\text{Conv}(\text{Linear}(x_{mid})))) \times \sigma(\text{Linear}(x_{in})) \quad (1)$$

Figure.1 shows the structure of the SSM,SSM is the state space model of Mamba [7], where A is the control matrix, B is the state matrix ,C is the output matrix and D is the command matrix is described in detail in Algorithm 1.

The Kan network calculation formula is as follows:

$$x_i = (\Phi_{L-1} \circ \Phi_{L-2} \circ \dots \circ \Phi_1 \circ \Phi_0) \text{Norm}(\text{Mamba}(x)) \quad (2)$$

$$\text{where } \Phi_L = \begin{bmatrix} \phi_{l,1,1}(\cdot) & \phi_{l,1,2}(\cdot) & \dots & \phi_{l,1,n_l}(\cdot) \\ \phi_{l,2,1}(\cdot) & \phi_{l,2,2}(\cdot) & \dots & \phi_{l,2,n_l}(\cdot) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{l,n_l+1,1}(\cdot) & \phi_{l,n_l+1,2}(\cdot) & \dots & \phi_{l,n_l+1,n_l}(\cdot) \end{bmatrix}$$

is a learnable parameter matrix, which we will describe in detail in the following subsections.

B. Mamba Replace Feed-Forward

The Mamba network is an efficient neural network architecture designed to accelerate the inference process of deep learning models,its design goal is to enable neural networks to execute at extremely high speed and low latency through efficient hardware acceleration and optimized data flow processing.The main function of the Feed Forward layer is to further extract and process the input data.By replacing the Feed Forward in the Transformer layer with a Mamba network can enhance the model's ability to understand and process the data.

The main component of the Mamba network is the SSM, the state space model, and its algorithmic process is shown in Algorithm 1 as follows:

Algorithm 1 SSM

Input: $x : (B, L, D)$

Output: $y : (B, L, D)$

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1  $A : (D, N) \leftarrow$  Initial Vector
2  $B : (D, N) \leftarrow$  Initial Vector
3  $C : (D, N) \leftarrow$  Initial Vector
4  $D : (D, N) \leftarrow$  Initial Vector
5  $s_0 : (B, N) \leftarrow$  Initial State
6 for  $t = 1$  to  $L$  do
7    $u_t \leftarrow x[:, t, :]$ 
8    $s_t \leftarrow A \cdot s_{t-1} + B \cdot u_t$ 
9    $y_t \leftarrow C \cdot s_t + D \cdot u_t$ 
10 return  $y$ 
```

In this SSM algorithm, the input sequence is processed through four core matrices A, B, C, D . First, the matrix A controls the state transition of the system, describing how the state at the previous moment affects the current state; The matrix B maps the current input to the state space, affecting the current state update. Subsequently, the current hidden state is transformed into the output through the matrix C . At the same time, the inputs directly affect the outputs through the matrix D . In this process, each time step u_t in the input sequence x is sequentially operated with the state update formula $s_t = A \cdot s_{t-1} + B \cdot u_t$ and the output formula $y_t = C \cdot s_t + D \cdot u_t$, the final output sequence y . Therefore, the model uses a recursive state update mechanism and flexible matrix operations, making it suitable for tasks that require long-term dependencies, such as sequence modeling and signal processing.

C. KAN Assist Projection

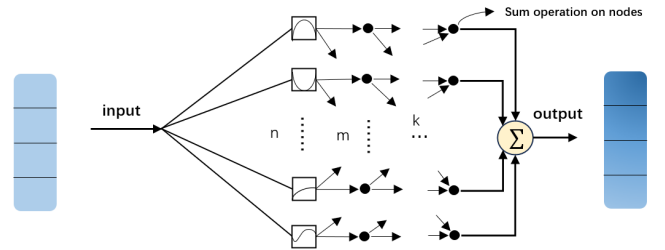


Fig. 2: KAN:reconstructing the input parameters

KAN is a novel fully-connected network structure that shares some similarities with multilayer perceptual machines (MLPs), but has significant design differences. In MLP, the activation function is fixed on the “neurons”; however, KAN places the activation function at the edges (a.k.a. “weights”), and the activation function is learnable. Therefore, instead of using a traditional linear weight matrix, KAN replaces each weight parameter with a learnable one-dimensional function

parameterized as a line. This structure makes KAN superior to MLP in terms of flexibility, which allows it to adapt better to complex tasks.

Therefore, we integrate this design of KAN into our network. Specifically, if f is a multivariate continuous function on a bounded domain, then in the design of Figure 2, f can be written as a finite combination of continuous functions of a single variable and a binary operation of addition. For this smooth function $f : [0, 1]^n \rightarrow R$, then there is the following equation:

$$output = \sum_{q=1}^k \Phi_q \left(\sum_{p=1}^n \phi_{q,p}(input) \right) \quad (3)$$

where both n and k are customized hyperparameters, n represents the number of learnable parameters, and k represents the number of learning layers. Defined by $\phi_{q,p} : [0, 1] \rightarrow R$; $\Phi_q : R \rightarrow R$,

As can be seen from Formula 3, their core is actually the operation of summing up single-variable functions. On this basis, other functions can be expressed by single-variable functions and their summation. Since all functions we want to learn are single-variable functions, we can parameterize each learnable one-dimensional function as a B-spline curve and use learnable coefficients with local B-spline basis functions. This method can effectively capture complex function characteristics.

The network's ability to reconstruct input parameters through continuous function approximation and B-spline application enables KAN to maintain smooth transitions between data points, thus alleviating the overfitting problem often encountered in deep neural networks. This flexibility also allows for more efficient learning and faster convergence during training.

IV. EXPERIMENTS

We evaluate our approach on three datasets: APAVA, ADFD, and PTB-XL.

A. Dataset introduction

The datasets used in this paper include APAVA [18], ADFD [19] and PTB-XL [20]. (1) APAVA has 23 subjects and 5967 16-channel multivariate EEG samples. Each sample consists of a one-second time series with 256 time stamps and a sampling rate of 256 Hz. The sample indicates whether the subject has Alzheimer's disease. (2) ADFD has 88 subjects and 69,762 19-channel multivariate EEG samples. Each sample is a one-second time series with 256 time stamps and a sampling frequency of 256 Hz. The sample is labeled to indicate whether the subject is healthy, has dementia, and has Alzheimer's disease. (3) The PTB-XL has a total of 17,596 subjects and 19,400 12-channel multivariate EEG samples. Each sample is a one-second time series with 250 time stamps and a sampling rate of 250 Hz. Each sample is labeled with five annotations indicating different cardiac conditions.

B. Implementation

In this experiment, we use six evaluation metrics: accuracy, precision (macro-mean), recall (macro-mean), F1 score (macro-mean), AUROC (macro-mean) and AUPRC (macro-mean). The training process is performed on a fixed training, validation, and test set, and the mean and standard deviation of the model are calculated using five random seeds (41-45). All experiments are run on NVIDIA RTX 4090 GPU servers, while we perform experiments on five servers of this model, and the final results are the mean of the servers run results.

For data augmentation in our approach, we choose to use masking, dithering, and scaling, three methods widely used in time series augmentation [1]. The batch sizes for datasets APAVA, ADFD, and PTB-XL are set to $\{32, 128, 128\}$, respectively. The number of training epochs is set to 100, and if the F1 score on the validation set does not improve, the training is stopped early after 10 epochs. Finally, we save the model with the highest F1 score on the validation set and evaluate it on the test set.

C. Results of Subject-Independent

Independent: In deep learning, "Independent" usually means that the features or variables are independent of each other and do not have interdependencies, which ensures that the model can effectively learn the independent contribution of each feature during the training process. In this experiment, the training set, validation set, and test set are divided according to the subjects. All subjects and their corresponding samples are assigned to the training set, validation set, and test set according to a pre-set ratio or subject ID. The samples of the same subject can only appear in one of the three sets.

results: From Table I, we can find that our method performs excellently on several datasets, especially on the APAVA and ADFD datasets, where it greatly outperforms the other comparative models in terms of accuracy, precision, recall, F1 scores and AUPRC, and it shows in dealing with the complex tasks showed better generalization ability and stability. In Table I, the later the experimental ranking indicates better results, and the first ranking indicates the best performance among all methods. Among all methods, our method has the best average ranking on all 6 metrics. It is also worth mentioning that the F1 score of our method on the ADFD dataset is significantly higher than other methods, and the other five metrics are also much better than other methods which results highlight the superior performance of our method.

D. Results of Subject-Dependent

Dependent: In deep learning, "dependent" data usually means that there is some kind of interdependence or association between data samples. This dependency means that the samples are not completely independent of each other and they may share some common characteristics or information. For "dependent" data, the training set, validation set, and test set may not be completely independent, and the dependency of samples will affect the training and evaluation of the model.

TABLE I: **Results of Subject-Independent Setup.** The training, validation, and test sets are distributed based on subjects according to a predetermined ratio/IDs. Results of the APAVA, ADFD, and PTB-XL datasets under this setup are presented here.

Datasets	Models	Accuracy	Precision	Recall	F1 score	AUROC	AUPRC
APAVA	Autoformer [1]	68.64±1.82	68.48±2.10	68.77±2.27	68.06±1.94	75.94±3.43	74.38±4.05
	Crossformer [1]	73.77±1.95	79.29±4.36	68.86±1.70	69.93±3.85	72.00±5.55	70.25±3.65
	FEDformer [1]	74.94±1.95	74.59±1.50	73.56±3.55	73.51±1.39	83.27±1.99	82.47±1.39
	Informer [1]	73.11±4.40	75.17±6.06	69.17±4.56	69.47±5.06	70.46±9.71	70.75±5.27
	iTransformer [1]	74.55±1.66	75.34±1.77	71.71±3.06	70.31±1.79	85.59±3.10	84.39±1.57
	MTST [1]	71.14±1.59	79.30±0.97	65.27±2.28	68.41±3.01	68.87±4.57	71.06±1.60
	Nonformer [1]	71.89±3.84	71.80±4.58	69.44±2.56	70.05±5.42	70.55±5.92	71.38±3.35
	PatchTST [1]	67.03±1.65	78.76±1.88	59.91±2.32	65.56±3.10	65.65±5.08	67.90±8.06
	Reformer [1]	78.70±2.00	82.50±3.95	75.00±1.61	76.41±1.32	73.94±1.41	73.31±6.40
	Transformer [1]	76.30±4.22	77.64±5.93	73.57±3.05	75.13±5.38	72.53±4.74	73.33±7.67
	Medformer [1]	78.74±0.64	81.11±0.84	75.40±0.60	76.31±0.71	83.20±0.49	83.66±0.92
	MamKanformer(ours)	80.35±0.57	83.40±1.59	77.02±0.55	78.05±0.41	83.56±2.11	84.40±1.31
ADFD	Autoformer [1]	45.25±1.48	43.67±1.94	42.96±2.03	42.59±1.85	61.02±1.82	43.10±2.30
	Crossformer [1]	50.45±2.31	45.57±1.63	45.88±1.82	45.50±1.70	66.45±2.03	48.33±2.05
	FEDformer [1]	46.30±0.59	46.05±0.76	44.22±1.38	43.91±1.37	62.62±1.75	46.11±1.44
	Informer [1]	48.45±1.96	46.54±1.68	46.06±1.84	45.74±1.38	65.87±1.27	47.60±1.30
	iTransformer [1]	52.60±1.59	46.79±1.27	47.28±1.29	46.79±1.13	67.26±1.16	49.53±1.21
	MTST [1]	45.60±2.03	44.70±1.33	45.05±1.30	44.31±1.74	62.50±0.81	45.16±0.85
	Nonformer [1]	49.95±1.05	47.71±0.97	47.46±1.50	46.96±1.35	66.23±1.37	47.33±1.78
	PatchTST [1]	44.37±0.95	42.40±1.13	42.06±1.48	41.97±1.37	60.08±1.50	42.49±1.79
	Reformer [1]	50.78±1.17	49.64±1.49	49.89±1.67	47.94±0.69	69.17±1.58	51.73±1.94
	Transformer [1]	50.47±2.14	49.13±1.83	48.01±1.53	48.09±1.59	67.93±1.59	48.93±2.02
	Medformer [1]	53.27±1.54	51.02±1.57	50.71±1.55	50.65±1.51	70.93±1.19	51.21±1.32
	MamKanformer(ours)	55.86±0.22	52.85±1.01	52.17±0.21	52.15±0.19	73.14±0.28	55.52±0.06
PTB-XL	Autoformer [1]	61.68±2.72	51.60±1.64	49.10±1.52	48.85±2.27	82.04±1.44	51.93±1.71
	Crossformer [1]	73.30±0.14	65.06±0.35	61.23±0.33	62.59±0.14	90.02±0.06	67.43±0.22
	FEDformer [1]	57.20±9.47	52.38±6.09	49.04±7.26	47.89±8.44	82.13±4.17	52.31±7.03
	Informer [1]	71.43±0.32	62.64±0.60	59.12±0.47	60.44±0.43	88.65±0.09	64.76±0.17
	iTransformer [1]	69.28±0.22	59.59±0.45	54.62±0.18	56.20±0.19	86.71±0.10	60.27±0.21
	MTST [1]	72.14±0.27	63.84±0.72	60.01±0.81	61.43±0.38	88.97±0.33	65.83±0.51
	Nonformer [1]	70.56±0.55	61.57±0.66	57.75±0.72	59.10±0.66	88.32±0.36	63.40±0.79
	PatchTST [1]	73.23±0.25	65.70±0.64	60.82±0.76	62.61±0.34	89.74±0.19	67.32±0.22
	Reformer [1]	71.72±0.43	63.12±1.02	59.20±0.75	60.69±0.18	88.80±0.24	64.72±0.47
	Transformer [1]	70.59±0.44	61.57±0.65	57.62±0.35	59.05±0.25	88.21±0.16	63.36±0.29
	Medformer [1]	72.87±0.23	64.14±0.42	60.60±0.46	62.02±0.37	89.66±0.13	66.39±0.22
	MamKanformer(ours)	73.57±0.41	64.67±0.84	60.95±0.27	62.83±0.49	90.15±0.05	67.52±0.40

TABLE II: **Results of Subject-Dependent Setup.** The training, validation, and test sets are split based on samples according to a predetermined ratio. Results of the ADFD dataset under this setup are presented here.

Datasets	Models	Accuracy	Precision	Recall	F1 score	AUROC	AUPRC
ADFD	Autoformer [1]	87.83±1.62	87.63±1.66	87.22±1.97	87.38±1.79	96.59±0.88	93.82±1.64
	Crossformer [1]	89.35±1.32	89.00±1.44	88.79±1.37	88.88±1.40	97.52±0.58	95.45±1.03
	FEDformer [1]	77.63±2.37	76.76±2.17	76.68±2.48	76.60±2.46	91.67±1.34	84.94±2.11
	Informer [1]	90.93±0.90	90.74±0.71	90.50±1.14	90.60±0.94	98.19±0.27	96.51±0.49
	iTransformer [1]	64.90±0.25	62.53±0.27	62.21±0.26	62.25±0.33	81.52±0.29	68.87±0.49
	MTST [1]	65.08±0.69	63.85±0.80	62.71±0.64	63.03±0.58	81.36±0.56	69.34±0.89
	Nonformer [1]	96.12±0.47	95.94±0.56	95.99±0.38	95.96±0.47	99.59±0.09	99.08±0.16
	PatchTST [1]	66.26±0.40	65.08±0.41	64.97±0.51	64.95±0.42	83.07±0.45	71.70±0.61
	Reformer [1]	91.51±1.75	91.15±1.79	91.65±1.56	91.14±1.83	98.85±0.35	97.88±0.60
	Transformer [1]	97.00±0.43	96.87±0.53	96.86±0.36	96.86±0.44	99.75±0.04	99.42±0.07
	Medformer [1]	97.62±0.34	97.53±0.33	97.48±0.40	97.50±0.36	99.83±0.05	99.62±0.12
	MamKanformer (ours)	98.65±0.20	98.64±0.17	98.50±0.15	98.59±0.18	99.92±0.03	99.84±0.08

Results: We evaluate the EEG dataset ADFD using the “dependent” setting in direct comparison with the “subject-independent” setting. The results are shown in Table II. Our method achieves the best results in all six evaluations, especially with a high F1 score of 98.59%. It is worth mentioning that the baseline methods Informer, Nonformer, Reformer, Transformer, and Medformer also performed well, with F1 scores above 90%.

E. Ablation Experiment

In this section, to evaluate the experimental effects of Mamba and Kan, we randomly choose a machine and perform ablation experiments on three datasets with two different setups: no Mamba and no Kan. Table III compares the full MamKanformer model with these two variants. On each dataset, the overall evaluation scores of the full MamKanformer model exceed the scores of the individual components, showing strong performance. We find that the No Mamba component improves scores more than No Kan on these datasets, but that No Kan is able to complement No Mamba

TABLE III: Ablation Experiment

Datasets	Models	Accuracy	Precision	Recall	F1 score	AUROC	AUPRC
APAVA	No Mamba	79.59	81.28	76.65	77.55	82.07	83.06
	No Kan	78.26	80.09	75.08	75.92	81.38	82.72
	MamKanformer	80.15	81.25	77.57	78.42	85.64	85.68
ADFD (InDependent)	No Mamba	51.03	48.76	48.96	48.83	68.36	48.80
	No Kan	53.86	52.75	51.01	51.04	71.98	52.47
	MamKanformer	55.82	53.05	52.77	52.45	73.04	55.51
ADFD (Dependent)	No Mamba	98.47	98.46	98.35	98.40	99.92	99.83
	No Kan	98.50	98.44	98.49	98.46	99.91	99.83
	MamKanformer	98.69	98.64	98.61	98.62	99.94	99.87
PTB-XL	No Mamba	73.10	64.77	60.58	62.27	89.93	66.93
	No Kan	73.25	64.79	60.22	61.92	90.13	66.72
	MamKanformer	73.17	64.83	60.70	62.36	90.19	67.15

to achieve unexpected levels of experimental results. Overall, these results illustrate the critical role of each component in our design.

V. CONCLUSION

In this paper, we propose a deep learning model MamKanformer for EEG signal classification. We modified Medformer by using Mamba to solve the problem of Feed Forward's difficulty in effectively modeling long-short-term dependencies, and then using the output of Kan as the input part of the model's projection to make the network's classification more effective. These mechanisms empower the model to undertake cross-channel patching involving multiple timestamps and multi-channel characteristics, execute multi-granularity embedding for the purpose of capturing features at diverse scales, and carry out two-stage multi-granularity self-attention operations to seize the features both within and between different granularities. Based on these processes, the model is capable of achieving a more precise fitting of the output results to the true values. Meanwhile the results obtained on three datasets are compared with 11 baselines, demonstrating the effectiveness and stability of our method and highlighting its potential for real world applications.

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REFERENCES

- [1] Y. Wang, N. Huang, T. Li, Y. Yan, and X. Zhang, "Medformer: A Multi-Granularity Patching Transformer for Medical Time-Series Classification," *arXiv preprint*, 2024, arXiv:2405.19363.
- [2] Chaoqi Yang, M Brandon Westover, and Jimeng Sun. Manydg: Many-domain generalization for healthcare applications. In The Eleventh International Conference on Learning Representations, 2023.
- [3] Qiao Xiao, Khuan Lee, Siti Aisah Mokhtar, Iskasyar Ismail, Ahmad Luqman bin Md Pauzi, Qiuxia Zhang, and Poh Ying Lim. Deep learning-based eeg arrhythmia classification: A systematic review. *Applied Sciences*, 13(8):4964, 2023.
- [4] Yingying Jiao, Yini Deng, Yun Luo, and Bao-Liang Lu. Driver sleepiness detection from eeg and eeg signals using gan and lstm networks. *Neurocomputing*, 408:100–111, 2020.
- [5] Vernon J Lawhern, Amelia J Solon, Nicholas R Waytowich, Stephen M Gordon, Chou P Hung, and Brent J Lance. Eegnet: a compact convolutional neural network for eeg-based brain-computer interfaces. *Journal of neural engineering*, 15(5):056013, 2018.
- [6] M. Bock, A. Hölzemann, M. Moeller, and K. Van Laerhoven, "Improving Deep Learning for HAR with Shallow LSTMs," in *2021 International Symposium on Wearable Computers*, ACM, Sep. 2021, doi: 10.1145/3460421.3480419.
- [7] A. Gu and T. Dao, "Mamba: Linear-Time Sequence Modeling with Selective State Spaces," *arXiv preprint*, 2023, arXiv:2312.00752.
- [8] S. Sun, W. Ren, J. Zhou, J. Gan, R. Wang, and X. Cao, "A Hybrid Transformer-Mamba Network for Single Image Deraining," *arXiv preprint*, 2024, arXiv:2409.00410.
- [9] O. Lieber, B. Lenz, H. Bata, G. Cohen, J. Osin, I. Dalmedigos, E. Safahi, S. Meiroum, Y. Belinkov, S. Shalev-Shwartz, O. Abend, R. Alon, T. Asida, A. Bergman, R. Glozman, M. Gokhman, A. Manevich, N. Ratner, N. Rozen, E. Shwartz, M. Zusman, and Y. Shoham, "Jamba: A Hybrid Transformer-Mamba Language Model," *arXiv preprint*, 2024, arXiv:2403.19887.
- [10] Z. Liu, Y. Wang, S. Vaidya, F. Ruehle, J. Halverson, M. Soljačić, T. Y. Hou, and M. Tegmark, "KAN: Kolmogorov-Arnold Networks," *arXiv preprint*, 2024, arXiv:2404.19756.
- [11] C. Dong, L. Zheng, and W. Chen, "Kolmogorov-Arnold Networks (KAN) for Time Series Classification and Robust Analysis," *arXiv preprint*, 2024, arXiv:2408.07314.
- [12] C. J. Vaca-Rubio, L. Blanco, R. Pereira, and M. Caus, "Kolmogorov-Arnold Networks (KANs) for Time Series Analysis," *arXiv preprint*, 2024, arXiv:2405.08790.
- [13] Y. Chen, Z. Zhu, S. Zhu, L. Qiu, B. Zou, F. Jia, Y. Zhu, C. Zhang, Z. Fang, F. Qin, J. Fan, C. Wang, Y. Gao, and G. Yu, "SCKansformer: Fine-Grained Classification of Bone Marrow Cells via Kansformer Backbone and Hierarchical Attention Mechanisms," *arXiv preprint*, 2024, arXiv:2406.09931.
- [14] A. Gu, K. Goel, and C. Ré, "Efficiently Modeling Long Sequences with Structured State Spaces," in *The International Conference on Learning Representations (ICLR)*, 2022.
- [15] Yong Liu, Tengge Hu, Haoran Zhang, Haixu Wu, Shiyu Wang, Lintao Ma, and Mingsheng Long. itransformer: Inverted transformers are effective for time series forecasting. *International Conference on Learning Representations*, 2024.
- [16] Haoyi Zhou, Shanghang Zhang, Jieqi Peng, Shuai Zhang, Jianxin Li, Hui Xiong, and Wancai Zhang. Informer: Beyond efficient transformer for long sequence time-series forecasting. In *Proceedings of the AAAI conference on artificial intelligence*, volume 35, pages 11106–11115, 2021.
- [17] Tian Zhou, Ziqing Ma, Qingsong Wen, Xue Wang, Liang Sun, and Rong Jin. FEDformer: Frequency enhanced decomposed transformer for long-term series forecasting. In *Proc. 39th International Conference on Machine Learning (ICML 2022)*, 2022b.
- [18] J Escudero, Daniel Abásolo, Roberto Hornero, Pedro Espino, and Miguel López. Analysis of electroencephalograms in alzheimer's disease patients with multiscale entropy. *Physiological measurement*, 27(11):1091, 2006.
- [19] Andreas Miltiadous, Emmanouil Gionanidis, Katerina D Tzimourta, Nikolaos Giannakeas, and Alexandros T Tzallas. Dice-net: a novel convolution-transformer architecture for alzheimer detection in eeg signals. *IEEE Access*, 2023.
- [20] Patrick Wagner, Nils Strodthoff, Ralf-Dieter Bousselet, Dieter Kreiseler, Fatima I Lunze, Wojciech Samek, and Tobias Schaeffter. Ptb-xl, a large publicly available electrocardiography dataset. *Scientific data*, 7(1):1–15, 2020.